

Thermal-Hydraulic Python Simulation of a Passive Emergency Heat Removal System (EHRS)

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Abstract

This study presents the development and validation of a steady-state numerical model for the Emergency Heat Removal System (EHRS), a passive safety feature designed for the IRIS Generation III+ reactor. The simulation is based on the experimental data and geometric configuration of the IES (IRIS EHRS Simulator) facility, located at the SIET laboratories in Piacenza and operated in collaboration with Politecnico di Milano. The system operates as a closed-loop, two-phase natural circulation circuit under sliding pressure conditions. A custom Python-based simulation code was developed to characterize the thermal-hydraulic performance, specifically focusing on the balance between gravitational driving forces and frictional pressure drops across the helical coil steam generator, the riser, and the condenser. The results demonstrate the model's capability to replicate the experimental trends observed at the SIET facility, providing a reliable tool for assessing passive heat removal during reactor transients.

1. Introduction

The safety strategies for Generation III+ and IV nuclear reactors, particularly Small Modular Reactors (SMRs), increasingly rely on Passive Safety Systems (PSS) to enhance plant reliability and simplify design architectures. Unlike traditional active systems, PSS utilize fundamental natural laws—such as gravity and natural circulation—to perform safety functions without the need for external power sources or mechanical actuation. By eliminating active components like pumps, these systems significantly reduce the probability of hardware failure, offering a robust solution for decay heat removal during accident scenarios such as Station Blackouts.

1.1. The IRIS Emergency Heat Removal System

A notable application of this philosophy is the Emergency Heat Removal System (EHRS) of the IRIS reactor (International Reactor Innovative and Secure). The IRIS EHRS is designed to transfer core decay heat and sensible heat from the primary coolant to an external heat sink, specifically the Refueling Water Storage Tank (RWST), via a natural circulation loop.

The system configuration differs from typical Pressurized Water Reactor (PWR) designs; it operates as a closed loop, two-phase flow system characterized by sliding pressure behavior. This implies that the loop pressure is thermodynamically coupled to the heat balance between the source and the sink, rather than being controlled by a pressurizer. The loop consists of a heat source (Steam Generator), a riser, a heat sink (Pool Condenser), and a downcomer. The density difference between the hot fluid in the riser and the cold condensate in the downcomer generates the necessary gravitational driving head to sustain the flow.

1.2. Experimental Validation: The IES Facility

To investigate the complex thermal-hydraulic phenomena inherent to this design—such as system stability at low pressures and the effects of non-condensable gases—the IRIS EHRS Simulator (IES) was constructed at the SIET laboratories in Piacenza, Italy.

As depicted in the facility documentation, the IES is a full-height experimental rig. This vertical elevation (≈ 20 m) is a critical design feature, ensuring that the geodetic head driving the natural circulation matches that of the full-scale reactor, thereby preserving the scaling of gravitational forces. The facility includes a helical coil steam generator and a pool condenser, simulating the exact heat transfer mechanisms of the reference design.

1.3. Computational Modeling Strategy

Modeling natural circulation loops presents significant challenges due to the low driving forces involved, where the balance between buoyancy and friction is delicate. Furthermore, the presence of two-phase flow introduces complexity regarding void fraction prediction and phase slippage.

This study focuses on the development and application of a steady-state computational model implemented in Python to simulate the thermal-hydraulic behavior of the EHRS. The code adopts a nodalization approach, discretizing the loop components—including the helical steam generator, adiabatic piping, and pool condenser—to solve mass, momentum, and energy conservation equations.

To accurately capture the physics of the two-phase regions, the model integrates the Drift Flux Model (specifically using Zuber-Findlay type correlations), which accounts for the relative veloc-

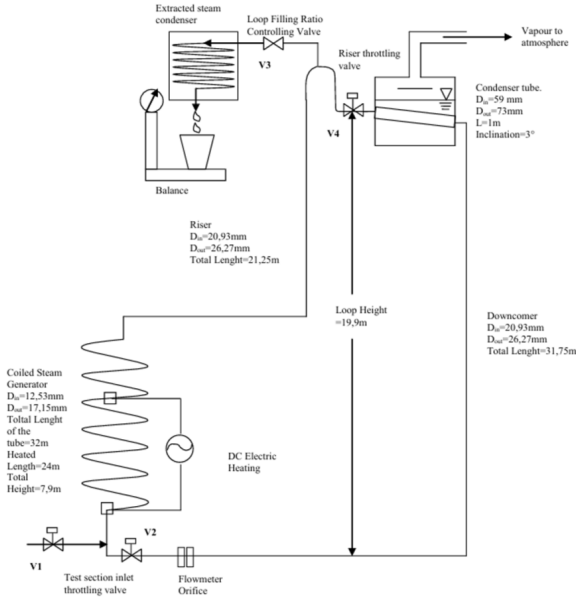


Figure 1. Scheme of the passive system

ity between liquid and vapor phases. This approach allows for a precise evaluation of the mixture density profile (ρ_m) along the loop height, which is the primary variable governing the system's mass flow rate and heat rejection capability.

The objective of this work is to provide a numerical tool capable of predicting the steady-state equilibrium of the system under various operating conditions, validating the code's logic against the known geometric and physical constraints of the IES facility.

2. General Modeling Approach

The numerical code developed for this study simulates the steady-state behavior of a natural circulation two-phase loop. The system is modeled as a closed thermodynamic cycle where the driving force is generated solely by density differences between the heated (riser) and cooled (downcomer) sections, without the aid of mechanical pumps.

The simulation adopts a one-dimensional (1D) finite volume approach. The fluid domain is discretized into a series of interconnected components (e.g., heated section, piping, condenser), which are further subdivided into spatial computational nodes. The solution is obtained by solving the conservation equations for mass, momentum, and energy across the entire loop, constrained by the fixed fluid mass inventory.

2.1. Fluid Properties and State Determination

The working fluid properties are calculated using the IAPWS-97 industrial formulation. At any given node, the thermodynamic state is defined by the local pressure P and specific enthalpy h . Based on these two state variables, the code determines the phase regime:

- **Single-phase Liquid:** If $h \leq h_{sat,liq}(P)$.
- **Two-phase Mixture:** If $h_{sat,liq}(P) < h < h_{sat,vap}(P)$.
- **Single-phase Vapor:** If $h \geq h_{sat,vap}(P)$.

The vapor quality x is calculated based on the enthalpy:

$$x = \frac{h - h_{sat,liq}}{h_{sat,vap} - h_{sat,liq}} \quad (1)$$

This phase detection logic dynamically selects the appropriate correlations for friction factors and heat transfer coefficients.

2.2. Conservation Equations

The governing equations are integrated over spatial steps dz along the flow path.

2.2.1. Conservation of Mass

Under steady-state conditions, the mass flow rate $\dot{m}$ is constant throughout the entire loop:

$$\frac{d\dot{m}}{dz} = 0 \Rightarrow \dot{m}_{in} = \dot{m}_{out} = \text{constant} \quad (2)$$

2.2.2. Conservation of Energy

The energy balance for a control volume accounts for heat input (source), heat extraction (sink), and environmental heat losses. The enthalpy change dh across a spatial step dz is given by:

$$\frac{dh}{dz} = \frac{1}{\dot{m}} \left(\frac{dQ_{source}}{dz} - \frac{dQ_{loss}}{dz} \right) \quad (3)$$

Where Q_{source} represents the electrical power input in the heated section, and Q_{loss} accounts for thermal dispersion to the ambient environment through the piping insulation or heat rejection in the condenser.

2.2.3. Conservation of Momentum

The pressure gradient along the flow path is the sum of three components: frictional losses, gravitational head, and accelerational pressure drop (due to density changes).

$$-\frac{dP}{dz} = \left(\frac{dP}{dz} \right)_{frict} + \left(\frac{dP}{dz} \right)_{grav} + \left(\frac{dP}{dz} \right)_{acc} \quad (4)$$

1. Frictional Term: Dependent on the flow regime. For single-phase flow, the Darcy-Weisbach equation is used. For two-phase flow, a two-phase multiplier Φ_{lo}^2 (Friedel correlation) is applied to the liquid-only pressure drop:

$$\left(\frac{dP}{dz} \right)_{frict} = \Phi_{lo}^2 \cdot \frac{f_{lo} G^2}{2\rho_l D} \quad (5)$$

2. Gravitational Term: This term drives the natural circulation. It depends on the local orientation of the pipe (angle θ with respect to the horizontal) and the mixture density ρ_{mix} :

$$\left(\frac{dP}{dz}\right)_{grav} = \rho_{mix} g \sin(\theta) \quad (6)$$

In two-phase regions, ρ_{mix} is calculated using a void fraction model (α), typically via Drift Flux correlations:

$$\rho_{mix} = \alpha \rho_v + (1 - \alpha) \rho_l \quad (7)$$

3. Accelerational Term: Significant in the heated section where phase change occurs (expansion). It is modeled using the change in momentum specific volume v_{mom} :

$$\Delta P_{acc} = G^2 (v_{mom,out} - v_{mom,in}) \quad (8)$$

where v_{mom} is defined as:

$$v_{mom} = \frac{x^2}{\alpha \rho_v} + \frac{(1-x)^2}{(1-\alpha) \rho_l} \quad (9)$$

2.3. System Constraints

The mathematical model is closed by imposing boundary conditions characteristic of a closed loop. The solver must find the unique combination of mass flow rate $\dot{m}$, initial pressure P_{start} , and initial enthalpy h_{start} such that:

- 1. Hydraulic Closure:** $\oint dP = 0$ (The pressure at the end of the loop matches the start).
- 2. Thermal Closure:** $\oint dh = 0$ (The fluid returns to the same thermodynamic state).
- 3. Mass Inventory:** The integral of fluid density over the total volume matches the charged mass M_{target} :

$$\int_{V_{sys}} \rho_{fluid} dV = M_{target} \quad (10)$$

3. Component Modeling

The system loop is discretized into specific components, each governed by distinct geometric parameters and conservation laws. The following subsections detail the mathematical formulations implemented for each component.

3.1. Global Constitutive Relations

Two specific models are applied globally across multiple components to determine heat losses and two-phase friction.

3.1.1. Variable Heat Loss Model

To match experimental observations, the overall heat transfer coefficient for losses to the ambient environment, U_{loss} , is modeled

as a function of the local fluid temperature T_{fluid} (in Kelvin). A linear interpolation is used, clamped between safety limits:

$$U_{calc} = 9.0 + 0.1 \cdot (T_{fluid} - 155 - 273.15) \quad (11)$$

The applied coefficient is constrained as:

$$U_{loss} = \max(8.0, \min(13.0, U_{calc})) \quad [\text{W/m}^2\text{K}] \quad (12)$$

The heat loss per node is calculated as $dQ_{loss} = U_{loss}(\pi D_{out} dL)(T_{fluid} - T_{amb})$.

3.1.2. Two-Phase Friction (Friedel Correlation)

In regions where $0 < x < 1$, the frictional pressure drop is calculated using the Friedel (1979) correlation. The two-phase multiplier Φ_{lo}^2 is defined as:

$$\Phi_{lo}^2 = E + \frac{3.24FH}{Fr^{0.045} We^{0.035}} \quad (13)$$

Where the dimensionless groups are computed using the mixture properties:

$$E = (1-x)^2 + x^2 \frac{\rho_l f_{go}}{\rho_v f_{lo}} \quad (14)$$

$$F = x^{0.78} (1-x)^{0.224} \quad (15)$$

$$H = \left(\frac{\rho_l}{\rho_v}\right)^{0.91} \left(\frac{\mu_v}{\mu_l}\right)^{0.19} \left(1 - \frac{\mu_v}{\mu_l}\right)^{0.7} \quad (16)$$

Here, $Fr = G^2 / (g D \rho_h^2)$ and $We = G^2 D / (\sigma \rho_h)$, with ρ_h being the homogeneous density.

3.2. Heated Section (Source)

The heated section is modeled as a helical coil. The model accounts for the centrifugal effects on flow and phase distribution.

1. Geometric Correction (Ito's Approach): To account for secondary flows (Dean vortices) in the helical geometry, the single-phase friction factors (f_{lo} , f_{go}) used in the Friedel correlation are enhanced by a factor Y :

$$Y = 1 + C_{helix} \sqrt{\frac{D_{in}}{2R_{coil}}} \quad (17)$$

where C_{helix} is a tuning parameter derived from the system filling ratio ($\approx 1.8 \cdot \text{Ratio}_{fill}$).

2. Void Fraction (Drift Flux Model): The void fraction α is determined using a Drift Flux model suitable for inclined/vertical flows. The distribution parameter C_0 and drift velocity V_{gj} are:

$$V_{gj} = 1.53 \left[\frac{\sigma g (\rho_l - \rho_v)}{\rho_l^2} \right]^{0.25} \quad (18)$$

$$\alpha = \frac{j_g}{1.2j_{tot} + V_{gj}} \quad (19)$$

This α is used to calculate the gravitational head ΔP_{grav} and the momentum acceleration ΔP_{acc} .

3. Energy Balance: The enthalpy rise is calculated by subtracting heat losses from the electrical power input Q_{el} :

$$h_{out} = h_{in} + \frac{1}{\dot{m}} \left(\frac{Q_{el}}{N_{nodes}} - dQ_{loss} \right) \quad (20)$$

3.3. Orifice Section

The orifice creates a localized pressure drop. The loss coefficient K_{orif} is modeled as a function of the Reynolds number (Re) to capture the transition from laminar to turbulent flow through the restriction:

$$K_{orif} = 38.38 \left[1 - \exp \left(-\frac{Re}{728.8} \right) \right] \quad (21)$$

The total pressure drop across the orifice is the sum of distributed friction, gravitational head, and the concentrated loss:

$$\Delta P_{total} = \Delta P_{dist} + \rho g \Delta z + K_{orif} \frac{G^2}{2\rho_{in}} \quad (22)$$

3.4. Condenser (Heat Sink)

The condenser is modeled as a tube submerged in a pool at temperature T_{pool} . The heat transfer is governed by a thermal resistance network:

$$dQ = \frac{T_{fluid} - T_{pool}}{R_{int} + R_{wall} + R_{ext}} \quad (23)$$

where $R = 1/(hA)$. The heat transfer coefficients (HTC) are determined as follows:

- **External (h_{ext}):** Modeled as a high-efficiency pool boiling process with a fixed coefficient $h_{pool} \approx 25,000 \text{ W/m}^2\text{K}$.
- **Internal Condensation (h_{int}):** For the two-phase condensing region, a Shah-type correlation is implemented. The single-phase liquid HTC (h_{LO}) is enhanced by a factor Z :

$$h_{2\phi} = h_{LO} \cdot Z \cdot \psi_{tuning} \quad (24)$$

The multiplier Z accounts for the wetting perimeter and interfacial shear:

$$Z = (1 - x)^{0.8} + \frac{3.8x^{0.76}(1 - x)^{0.04}}{(P_{red})^{0.38}} \quad (25)$$

where $P_{red} = P/P_{crit}$. A tuning factor $\psi_{tuning} \approx 12.0$ is applied to match the high heat flux capability observed in the thin-film condensation regime.

3.5. Passive Piping and Elbows

The risers, downcomers, and siphon are modeled primarily as adiabatic pipes with thermal dispersion (U_{loss}).

1. Elbows: Local pressure losses in elbows are calculated using a

geometric K -factor dependent on the radius of curvature R_{curv} :

$$K_{bend} = \left(0.10 + 1.85 \left(\frac{D}{2R_{curv}} \right)^{3.5} \right) \cdot \frac{\theta_{bend}}{90^\circ} \quad (26)$$

This allows the model to distinguish between sharp 90° turns and smoother bends.

2. Downcomers: In the downcomers, the gravitational term is positive (pressure recovery). The code checks specifically for flow regime; while typically single-phase liquid, if vapor entrainment occurs ($x > 0$), the Friedel multiplier is automatically engaged to correctly predict the increased resistance.

4. Loop Solver and Solution Algorithm

The steady-state operating point of the passive loop is determined by solving a non-linear system of equations. Since the loop is closed, the fluid properties at the outlet of the last component must strictly match the properties at the inlet of the first component. Additionally, the total mass of the fluid in the system is fixed by the initial filling ratio.

4.1. State Vector

The solver iterates on a state vector $\mathbf{X}$ containing three independent variables that fully define the boundary conditions of the cycle:

$$\mathbf{X} = \begin{bmatrix} \dot{m} \\ P_{in} \\ h_{in} \end{bmatrix} \quad (27)$$

where $\dot{m}$ is the mass flow rate (kg/s), P_{in} is the pressure (Pa) at the inlet of the test section, and h_{in} is the specific enthalpy (J/kg) at the same location.

4.2. Objective Function and Residuals

The simulation executes the component models sequentially around the loop (Heated Section $\rightarrow$ Risers $\rightarrow$ Condenser $\rightarrow$ Downcomers). At the end of the calculation chain, a residual vector $\mathbf{R}(\mathbf{X})$ is computed. The problem is formulated as finding $\mathbf{X}$ such that $\mathbf{R}(\mathbf{X}) \approx \mathbf{0}$.

The residual vector consists of three closure errors:

- 1. Hydraulic Closure (ε_P):** The pressure recovered in the downcomers must exactly balance the pressure drops in the risers and components.
- 2. Thermal Closure (ε_h):** The enthalpy at the end of the loop must match the start, ensuring energy conservation.
- 3. Mass Inventory Closure (ε_M):** The sum of the fluid mass in every component must equal the target system mass.

The residuals are normalized and weighted to improve numerical stability:

$$\mathbf{R}(\mathbf{X}) = \begin{bmatrix} \frac{P_{out,final} - P_{in}}{1000} \\ \frac{h_{out,final} - h_{in}}{1000} \\ 10 \cdot (\sum_i \rho_i V_i - M_{target}) \end{bmatrix} \quad (28)$$

The target mass M_{target} is defined by the system volume V_{sys} , the reference density ρ_{ref} (at cold conditions), and the filling ratio ϕ :

$$M_{target} = V_{sys} \cdot \phi \cdot \rho_{ref} \quad (29)$$

4.3. Numerical Optimization Method

The system is solved using the Trust Region Reflective algorithm (via the `scipy.optimize.least_squares` library). This method minimizes the sum of squared residuals:

$$\min_{\mathbf{X}} \frac{1}{2} \|\mathbf{R}(\mathbf{X})\|^2 \quad (30)$$

subject to physical bounds:

$$\mathbf{L} \leq \mathbf{X} \leq \mathbf{U} \quad (31)$$

Typical bounds used are $0.001 < \dot{m} < 1.0$ kg/s and $1 < P < 50$ bar, ensuring the solver avoids unphysical regions (e.g., negative flow or vacuum pressures) during iteration. The convergence criteria are set to a function tolerance of 10^{-6} .

5. Conclusion

The computational model developed in this study demonstrates a robust capability to predict the steady-state performance of the passive two-phase loop, functioning as an Emergency Heat Removal System (EHRS). By implementing a 1D finite volume approach and a non-linear solver strategy in Python, the fundamental physics of the two-phase natural circulation cycle have been effectively investigated.

To validate the model, the numerical predictions were compared against the available experimental data obtained from the facility. Table 1 summarizes the experimental steady-state values for the key operating parameters—specifically the mass flow rate, inlet pressure, fluid temperature, and enthalpy—under different configurations of supplied thermal power and filling ratio (FR).

Table 1. Experimental steady-state results for different operating conditions.

Power [kW]	FR [-]	$\dot{m}$ [kg/s]	P_{in} [kPa]	T_{in} [K]	h_{in} [kJ/kg]
34	0.49	0.0420	1628.69	437.61	695.53
44	0.49	0.0490	2806.23	459.72	792.93
34	0.31	0.0429	1259.45	457.62	783.75

The corresponding numerical results obtained from the developed Python computational model are presented in Table 2.

Comparing the two sets of data, the numerical results show a very close agreement with the experimental trends for the cases operating at a filling ratio of 0.49. The model accurately captures the system's response to the variation in thermal power from 34 kW to 44 kW, showing consistent trends in inlet pressure, temperature, and enthalpy. However, it is important to note a general, slight underestimation of the mass flow rate ($\dot{m}$) across

Table 2. Numerical results obtained from the Python computational model.

Power [kW]	FR [-]	$\dot{m}$ [kg/s]	P_{in} [kPa]	T_{in} [K]	h_{in} [kJ/kg]
34	0.49	0.0364	1658.37	439.06	701.86
44	0.49	0.0454	2862.35	461.88	802.69
34	0.31	0.0272	1551.64	444.64	726.12

all three simulated cases compared to the experimental reference. This consistent offset suggests a potential overestimation of the frictional pressure drops within the two-phase regions of the loop by the empirical correlations adopted.

For the operating condition with a lower filling ratio (34 kW, 0.31 FR), larger discrepancies are observed between the numerical output and the experimental data. While the model correctly predicts a reduction in mass flow rate compared to the 0.49 FR cases, the overall thermodynamic state deviates more significantly. Nevertheless, this discrepancy is not entirely attributable to the simulation tool; it must be highlighted that the reference theoretical/experimental dataset for the 0.31 FR condition inherently presents higher uncertainties and potential consistency issues, making a direct and precise validation more challenging in this specific regime.

Despite the satisfactory overall accuracy of the simulation, the residual deviations highlight the inherent limitations of standard 1D constitutive laws. The empirical correlations used for two-phase frictional pressure drops and heat transfer coefficients may not fully account for the specific geometric features of the experimental facility, such as the helical coil effects or the complex flow regimes in the condenser. Future work should focus on calibrating these constitutive relations against a broader and more reliable dataset, particularly for low filling ratios, to minimize residual errors and enhance the model's reliability across a wider range of operating conditions.